AI CW

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May 2024

1 Introduction

The penguins dataset features measurements of various aspects of 3 species of penguin, ('species' will also be described as 'breed/breeds' in the code, when there is a necessity to distinguish between a singular and plural noun). The dataset records each penguins: sex, bill length, bill depth, body mass, flipper length, species and the island the penguin was found on.

1.1 Cleaning

Firstly, it was necessary to address N/A values within the dataset. There were 11 N/A values. Each of these penguins had an N/A value for the penguins sex. Interpolating a discrete value like sex is challenging. The dataset has sufficient fully labelled data, to exclude these values. As an extension it may be possible to use logistic regression to predict a penguins gender, based on its other characteristics, however this would introduce uncertainty, and is beyond the scope of this project.

The sex variable of the remaining penguins is discrete and as processing technique, was converted to a float value "is_female". With 1 indicating that a penguin is female, and 0 indicating that a penguin is male.

The island the penguin was found on is not included in this report. Both Gentoo and Chinstrap penguins were found exclusively on one island, so this variable offers little insights beyond the species. It is considered an extension to include this variable in predictions in a way that is meaningful.

For each species of penguin, a train/test/validate split of 70/15/15 was used. This was done by species, to ensure in each set there was sufficient representation from each species. This is important when predicting species, as if one species was particularly underrepresented in the test data, it would be difficult to accurately quantify how well a model performs.

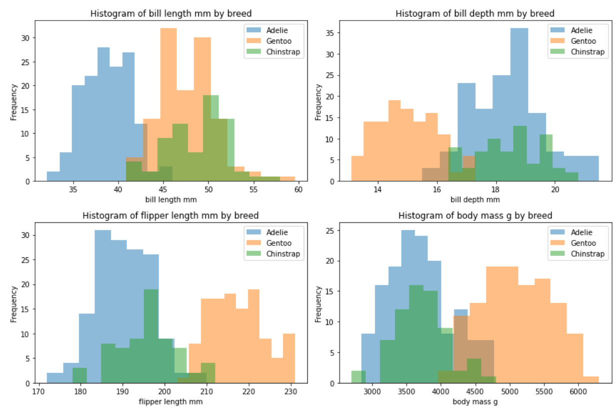

Figure 1: Distribution of penguin measurements

1.2 Exploration

Histograms can be used to show the distribution of various continuous features of the penguins within the dataset. This is shown in Figure 1. This display shows clear distinctions between measurements of Adelie and Gentoo penguins, with little overlap. Chinstrap penguins seem to have a more sparse distribution across variables, and seem to have a greater overlap with the other two species. It seems that each variable, for each penguin is normally distributed. This has not been statistically tested, but could be as an extension.

Figure 1 shows that characteristics associated with

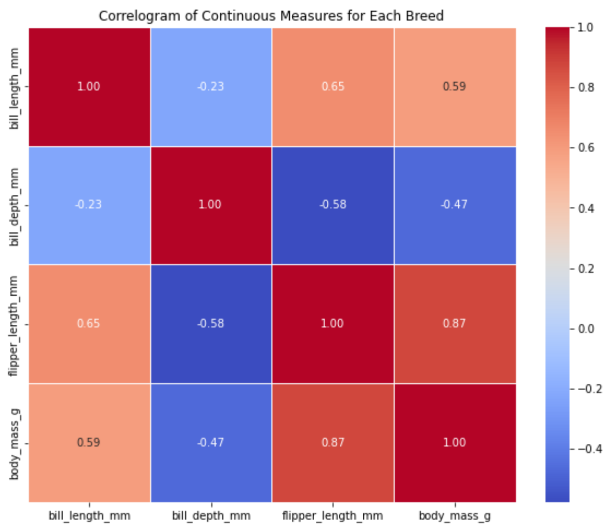

Figure 2: Correlation between variables

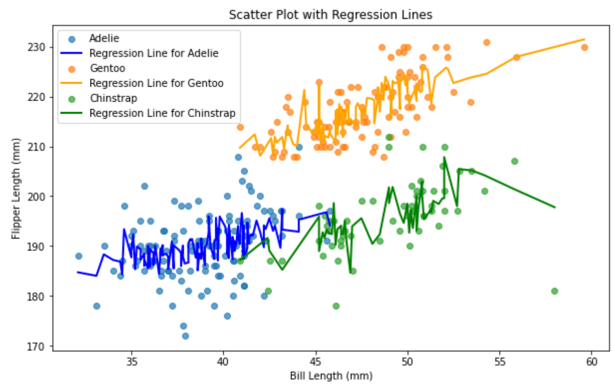

Figure 3: Scatter Plot showing Linear Regression Predictions

Gentoo penguins tend to include: larger body mass, longer flipper length and shorter bill depth, while Adelie penguins have characteristics showing shorter bill length, shorter flipper length and deeper bill depth. The characteristics of Chinstrap penguins are less precise and more variable.

Representing sex in a similar format would be inappropriate as there are only two values each species can take. Consequentially it is justified to calculate the proportion of sex between each species, respectfully: 50.0%, 48.74% and 50.0% of Adelie, Gentoo and Chinstrap penguins are female, which is a fairly proportional split.

Figure 2 shows a correlogram depicting how different continuous variables correlate with each other. The strongest correlation (or anti-correlation) between any two variables is between flipper length and body mass with correlation coefficient of 0.87. Consequentially these values are used in for visualising the regression model, to display trends best.

The variable with highest absolute correlation between variables is the flipper length, thus, it is used as the predicted variable in the linear regression model.

2 Regression

To predict flipper length according to the other variables, a multivariate linear regression model (with lasso penalisation parameter α), is constructed for each breed of penguin as follows:

$$\text{For } \gamma \in [\text{Adelie, Gentoo, Chinstrap}]:$$

$$\text{Flipperlength}_\gamma = \beta_\gamma(\alpha_\gamma)\mathbf{X}_\gamma + \epsilon_\gamma$$

Where the subscript γ indicates the different values by breed. The regression coefficient is $\beta(\alpha)$, which the model aims to optimise (with constraining lasso parameter α). This lasso parameter is used to scale down variables with less importance in predicting flipper length. The design matrix: $\mathbf{X}$ contains information for each variable for each penguin in each breed. The error term is represented by ϵ.

The α_γ parameter is selected according to grid search, the value for α_γ that minimises each Residual Mean Squared Error: RMSE_γ for the validation data is selected. The table of values is shown in Table 5 in the appendix for completeness.

The predictions made according to each multivariate linear regression model are shown in 3. The line of predicted values is shown too, this shows the predicted value for each penguin, according to the length of its bill. The Regression line not being straight is due to the fact that the design matrix and coefficients are multidimensional - these predictions are

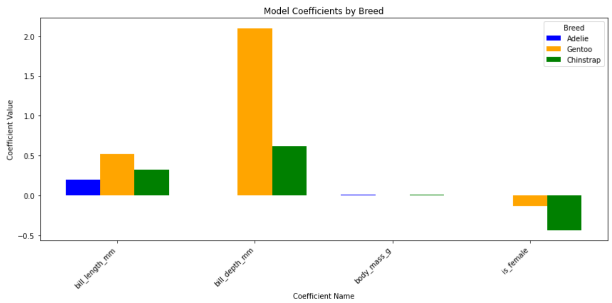

Figure 4: Coefficient Values by Model

not drawn linearly from bill length. It is too challenging to visualise all four dimensions of prediction data.

The values for each β coefficient reflect the weight of how much each variable contributes in predictions. These values are shown in 4. It is apparent that body mass has little weight in predicting flipper length for all three models, this is surprising as this variable has high correlation with flipper length, and correlation between the two variables is apparent in the scatter plot. The explanation for this might be that the correlation between body mass and the other variables give more importance to the other variables in predictions.

For Gentoo and Chinstrap penguins, the bill depth contributes most in predicting flipper length. The female variable shows that female Gentoo and Chinstrap penguins have shorter flippers (due to the negative coefficient associated with being female) where as there is no effect for Adelie penguins, for which the only variable of significant interest is the bill length.

The RMSE of each model on the respective test data are respectfully as follows: $4.89, 4.71$ and 4.55 for Adelie, Gentoo and Chinstrap penguins. This shows the models did a good and similar prediction of flipper length.

3 Classification

The next task involves predicting the class (species) of each penguin from the other variables in the dataset. The two following models were used to achieve this.

The accuracy is chosen as the metric of evaluation, due to its interpretability. The trade-off between the impact of false positives and false negatives has no real world implications.

A baseline model was used to predict the class according to which class was the majority. This is obviously a poor choice of algorithm, and gave an accuracy of 43.14

3.1 K Nearest Neighbours

A K-Nearest Neighbours (KNN) algorithm was used to predict penguin species. This algorithm works by selecting the label of the k closest (in this model, closest is defined in euclidean space) other data points.

Before initiating a model, it is important to scale each coefficient, this is to ensure that the euclidean space between data is on the same scale, otherwise one variable with a small range may dominate predictions.

The value of k is chosen according to grid search again. Values of k are chosen from one to seven. Calculating the accuracy on the validation data shows that values of k of $1, 3$ and 5 give 100% accuracy. Between these three values, 1 was eliminated, as such a low number of neighbours would provide for a model that isn't very robust and might under-fit. $k = 5$ may prove too large, and may result in over-fitting, consequentially k is chosen as 3. Whilst this accuracy is great, it raises concerns about the problem setting. It is possible that the train/test/validate split is too disproportionate, in favour of the training data, meaning predictions made are incredibly accurate, because the validation and test sets don't have enough diversity within them.

3.2 Random Forest

The random forest model uses a series of decision trees to predict class label.

A decision tree consists of nodes and splits. The tree starts with the root node with every data point, then at each level, nodes are split according to certain criteria, one criterion being the purity of the node (the proportion of the majority class found at this node). At the final nodes, data is classified is drawn according to which class is the majority in the training data found at the given terminal node.

The gini index is (among others) used as a measure of purity (the gini index is chosen in this report due to its simplicity). The gini index is calculated according to the proportion of the data that falls into each split, measuring probabilities rather than entropy.

The random forest is a combination of decision trees, each trained on a different random subset of the data. The random forest predicts the class that is chosen by most of the decision trees.

Other selections necessary for the model are:

- Number of decision trees in the random forest.
- The maximum depth of the tree (how many splits is too many).
- The minimum number of data points of a class requited to justify a to split.
- The minimum number of data points required to be at a terminal node.

These parameters are chosen according to grid search, on which combination maximises the accuracy on the validation data. The table of values is not displayed as it is less succinct than the table of values in lasso linear regression.

The optimal hyper parameters were found to be as follows:

- Number of decision trees in the random forest (10).
- The maximum depth of the tree (none).
- The minimum number of data points of a class requited to justify a to split (2).
- The minimum number of data points required to be at a terminal node (1).

3.3 Comparison

To compare performance between models, confusion matrices are used. Confusion matrices show visually how many of and how each class was correctly and

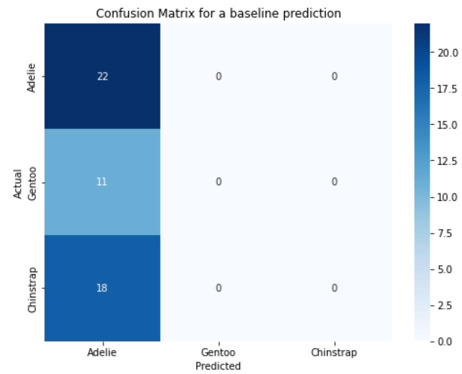

Figure 5: Baseline test performance

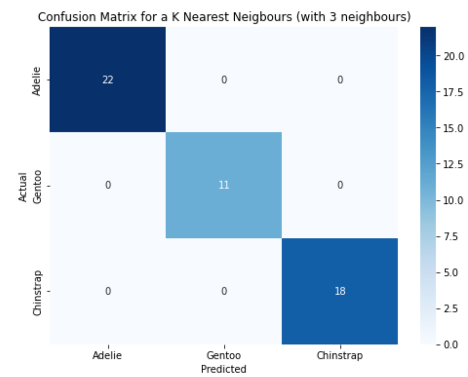

Figure 6: K-Nearest Neighbours test performance

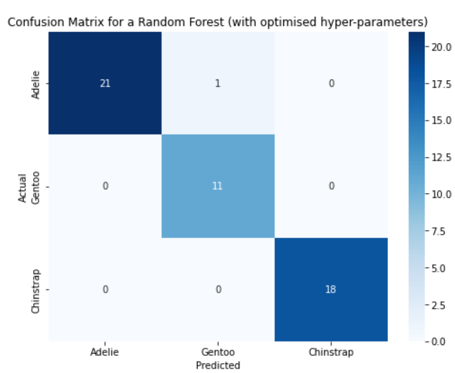

Figure 7: Random forest test performance

incorrectly classified, correct labels being on the diagonal.

It is clear to see in Figure 5 that the baseline model performs poorly on the test data, as expected. In figures 6, 7, the KNN algorithm and Random Forest have similar performance, with respective test accuracies: $100\%, 98.04$ (which are very high). The random forest performed very well, however predicted one Adelie penguin as Gentoo. This is surprising as Figure 1 showed that there were more distinct differences between the two variables, however it is possible that this resulted in stricter boundaries being drawn, causing outlierious values to be misclassified.

In conclusion, it is apparent that KNN outperforms random forests and baseline models, in predicting penguin species. This is supported by reference [1] which states that KNN algorithms outperform random forests due to their simplicity. This source talks about using models for data on forests, however the results can extend to data pertaining to species of penguins.

4 Ethics

It is important to consider the ethical implications of Artificial Intelligence (AI). The penguins dataset is a standard and commonly used dataset with limited ethical implications. One might say with the 'sex' variable it is important that penguin being male, as a variable of interest re-enforces anti-female bias. Consequentially the variable 'is_female' is used. This has a very minor ethical implication, however. Examples of more powerful ethical issues may include the following.

4.1 Amplification of Biases

Training AI algorithms on datasets that reflect the diversity of the human race is essential. Reference [2] explains how image recognition software performs worse when classifying dark skinned people. This can be largely due to the bias in training datasets that reflect real biases in society, particularly the bias that neglects people with darker skin, in lieu of people with lighter skin.

A specific example of this kind of bias having a negative societal effect, is the incident whereby Google images misclassified a black couple as "gorillas" [3]. This is an offensive racist trope that should never have happened, but it did due to poor training and poor testing from google.

In order to combat this bias it is crucial to train algorithms on data that capture real life diversity, with sufficient representation from social groups that face discrimination. Datasets might include sufficient representation from people of all races, economic backgrounds, genders, etc. Microsoft and IBM have since committed to diversifying training datasets [2].

It is also important to test the performance of these algorithms on the classification in potentially sensitive situations. This testing should ensure that there are no offensive projections of the data. This could be achieved through real life piloting, to ensure that predictions are not constricted by the original data.

4.2 Protection of Data

Data protection is another important facet to be considered in Data Science. An example that highlights a failure of this is the 2006 AOL data breach [4], whereby AOL released data on 20 million search queries of $65,000$ of its users. The data was anonymized to protect user identity, however, identities were still deducible due to personal information being preserved like location specific information.

A means of preventing occurrences like this occurring is through k-anonymity [5]. This process is used to ensure that no one user can be uniquely identified by their information. In fact, for each user there must be at least $(k-1)$ other users with the same information. This method is susceptible to re-identification attacks through cross referencing with other datasets (for example census data or voter registration information).

References

[1] Bosc, Laurence. *Assessment of Methods for Estimating Forest Stand Age from Field and LiDAR Data.* Forestry, April 2021. Available

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[2] Simone McCarthy. *The unseen Black faces of AI algorithms*. Nature, October 2022. Available at: https://www.nature.com/articles/d41586-022-03050-7.

[3] AI Incident Database. *Images of Black People Labeled as Gorillas*. Accessed on [Insert Date of Access]. Available at: https://incidentdatabase.ai/cite/16

[4] Michael Barbaro and Tom Zeller Jr. *A Face Is Exposed for AOL Searcher No. 4417749*. The New York Times, August 2006. Available at: https://www.nytimes.com/2006/08/09/technology/09aol.html.

[5] Charu C. Aggarwal and Philip S. Yu. *Outlier Detection for High Dimensional Data*. World Scientific, 2002. Available at: https://www.worldscientific.com/doi/abs/10.1142/S0218488502001648.

5 Appendix/Note

I used the article document class with 2 columns to best show figures with limited white space. I hope this is acceptable.

$\alpha_1, \alpha_2, \alpha_3$	**RMSE Adelie**	**RMSE Gentoo**	**RMSE Chinstrap**
(0.01, 0.01, 0.01)	5.017	4.708	4.802
(0.05, 0.05, 0.05)	4.987	4.719	4.763
(0.3, 0.3, 0.3)	4.919	4.726	4.589
(0.4, 0.4, 0.4)	4.901	4.732	4.556
(0.5, 0.5, 0.5)	4.892	4.741	4.546

Table 1: RMSE values for different choices of α (chosen values in red).